United International **University**

School of Science and Engineering

Trimester: Summer-2025

Course Title: Calculus and Linear Algebra

Course Code: MATH 2183, Assignment-2

Answer all the questions

1. (a) Given that,

$$\begin{gathered} \mathbf{4x - 8y + 4z = 0} \\ \mathbf{2y - 8z = 8} \\ \mathbf{-4x + 5y + 9z = -9} \end{gathered}$$

i) Find the inverse of A by using cofactor method and Inversion Algorithm method, hence solve the above system of linear equations.

ii) Find the unknown variables $x, y\ and\ z$ by using Cramer's rule.

(b) Also solve the system by Forward and Back substitution method

Solve $\boldsymbol{y'' + 8y' + 16y = \pi e^{2x} x^3 + Sin\, 2x + e^{-5x} + e^{x} \sin\ 2x}$

2. Find eigenvalues and eigenvectors of the Matrix $\boldsymbol{A} = \begin{bmatrix} \mathbf{2} & \mathbf{0} \\ \mathbf{1} & \mathbf{2} \end{bmatrix}$. Find the algebraic and geometric multiplicities .Also graph the Eigen spaces in $\boldsymbol{xy}$ −coordinates.

3. Given $\boldsymbol{B} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{2} & \mathbf{3} & \mathbf{1} \end{bmatrix}, \boldsymbol{A} = \begin{bmatrix} \mathbf{1} & \mathbf{2} \\ \mathbf{2} & \mathbf{0} \end{bmatrix}, \boldsymbol{C} = \begin{bmatrix} \mathbf{1} & \mathbf{2} \\ \mathbf{3} & \mathbf{0} \end{bmatrix}$

(a). Find $\boldsymbol{C_2(AB)}$, $\boldsymbol{r_1(AB)}$ $\boldsymbol{tr(B^{-2})}$, $\boldsymbol{det(C)}$**and** $\boldsymbol{det(B)}$

(b)

i) Find a matrix $\boldsymbol{X}$, Such that $\boldsymbol{(B + X)^T = C^T + A^T}$

ii) Find $\boldsymbol{p(A)}$ **for** $\boldsymbol{p(x) = x^2 + 2x + 3}$

iii) **Find $\boldsymbol{D}$ if** $\boldsymbol{(9D)^{-1} = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}}$

P.T.O

4.

In a track event between **Schools A and B**, students earned positions from **1st to 4th place**.

The number of places each school gained is shown below:

	1st	2nd	3rd	4th
School A	3	5	4	6
School B	5	3	6	4

Points for 1st, 2nd, 3rd, and 4th are **6, 4, 2, 0** respectively.

Let

$$A = \begin{pmatrix} 3 & 5 & 4 & 6 \\ 5 & 3 & 6 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 6 \\ 4 \\ 2 \\ 0 \end{pmatrix}$$

(i) What does each column of A represent?

(ii) Find AB.

(iii) Which school scored higher?

(iv) If the new points are 5, 3, 2, 1, would that change the result?